

Shangyu Andrew Xu

☎ (203) 909-8168 | ✉ andrew.xu@yale.edu | in [sandrewxu](#) | 🌐 [sandrewxu](#)

EDUCATION

Yale University

Aug 2023 – May 2027 (Expected)

Bachelor of Science in Computer Science

GPA: 3.97/4.0

Coursework

Statistics/ML: Statistical ML, Probabilistic ML, Probability Theory, Theory of Statistics, Statistical Methods

CS: Building AI Infra Systems, Natural Language Processing, Systems Programming, Deep Learning on Graphs, Spectral Graph Theory, Algorithms, Full Stack Web Development

Math: Honors Linear Algebra, Honors Analysis, Abstract Algebra, Vector Analysis, Discrete Math

Activities & Awards: YC AI Startup School, 5x AIME Qualifier, USAPhO Qualifier, USNCO Qualifier, Doox of Yale A Cappella, Yale Glee Club, Yale STEM Tour Guide, Yale College Democrats, The Yale Politic

EXPERIENCE

Incoming Quantitative Trading Intern (Summer 2026)

Aug 2025 – Present

Jane Street

New York, NY

NLP Researcher

May 2025 – Present

Yale Natural Language Processing Lab

New Haven, CT

- Designing and training coarse-to-fine diffusion language models under Dr. Arman Cohan (2025 Google Research Scholar) and Neil Band (Stanford NLP)
- Productionizing Self-Distillation Policy Optimization to multi-node GPU clusters under Dr. Y. Richard Yang

Head Teaching Assistant

Aug 2024 – May 2025

CS223: Data Structures and Programming Techniques, Yale University

New Haven, CT

- Delivered weekly section material on data structures and memory management in C for a 200-student class
- Developed 350+ public and private test cases for 6 brand-new coding projects in Fall 2024

Machine Learning Engineer Intern

May 2024 – Aug 2024

ActualFood

Remote

- Developed a full-stack iOS application in Swift, integrating a real-time CNN model for gesture recognition
- Designed and trained a CNN in PyTorch (inspired by MobileNetV4) for real-time hand gesture classification
- Quantized the CNN, reducing on-device inference time by 30% and model size by 50% on iPad devices

PUBLICATIONS

Centralized Selection with Preferences in the Presence of Biases | *ICML 2024*

Nov 2023 – Jul 2024

[Paper](#) | [Slides](#) | [Code](#)

- Co-first author and presenter; authored with Dr. Nisheeth Vishnoi, Dr. Elisa Celis, and Dr. Amit Kumar
- Researched the mathematical optimization of a centralized selection algorithm under unknown bias factors
- Independently created 40+ simulations testing the robustness of core theorem under non-ideal conditions

PROJECTS

Priority-Guided Masking for Diffusion Language Models | *PyTorch, HPC*

Mar 2025 - May 2025

[Paper](#) | [Slides](#) | [Code](#)

- Final project for CPSC 477: Natural Language Processing, winning the Best Project award in a graduate class
- Fine-tuned LLaDA-8B, a state-of-the-art diffusion language model, to prioritize the generation of important tokens first, resulting in a model that is able to perform better in less inference iterations

TECHNICAL SKILLS

Languages: Python, C, Rust, C++, Swift, JavaScript, HTML, CSS

Machine Learning Tools: PyTorch, TensorFlow, Hugging Face, Scikit-learn, NumPy, Pandas, Matplotlib

Developer Tools: Git, VS Code, Xcode, Linux/Unix, SLURM/HPC, React.js